Analytical Chemistry Cumulative Exam
December 4, 2009
Electrochemistry and Using the Nernst Equation
Equations and constants available at end of cume.

1. (20 pts) Consider a redox reaction between reactants HPO_3^{2-} and MnO_4^- that also involves the reactant OH-, and generates the products PO_4^{3-} and MnO_4^{2-} and possibly other substances.

(a) (8 pts.) Write out the overall balanced chemical equation? Show work

These questions will help you balance the equation above:
(b) (2 pt) What is the oxidation state of the P in HPO_3^{2-}? _____
(c) (2 pt) What is the oxidation state of the Mn in MnO_4^-? _____
(d) (2 pt) How many electrons are involved in converting one HPO_3^{2-} to one PO_4^{3-} ? _____
(e) (2 pt) How many electrons are involved in converting one MnO_4^- to one MnO_4^{2-} ? _____
(f) (2 pt) Which of the reactant species starts out in the reduced form? _____
(g) (2 pt) Which reactant species is the reductant? _____

2. (40 points) The following half reactions are important for hydrogen-based fuel cells. For certain concentrations and at 25 °C, the half-reaction potentials are given to you below, written as reductions (Note: "=" designates arrows in forward and reverse directions):

$2\ H^+ + 2\ e^- = H_2(g)$
(Where H_2 is at a partial pressure of 1 atm, and $[H+]=a^{H+} = 1$ M, then $E_{H+/H2} = 0.00$ V vs. NHE)

$O_2(g) + 4\ e^- + 4\ H^+ = 2\ H_2O$
(Where O_2 is at a partial pressure of 1 atm, H_2O is at a partial pressure of 1 atm, and $[H+]=a^{H+}$ is at 1 M, then $E_{O2/H2O} = +1.229$ V vs. NHE)

(a) (6 points) What are the E° values (under standard conditions) for each of these half reactions? Explain.
(b) (5 points) What is an "NHE"? Explain.
(c) (10 points) In order to produce a galvanic cell, which half reaction must be taking place at the anode? Also, what is the voltage of this cell when it is galvanic? Show work.
(d) (10 points) What would be the new voltage of the galvanic cell if the partial pressure of H_2O were decreased to 0.1 atm? Show work.
(e) (9 points) Name one big problem that needs to be overcome so that hydrogen fuel cells can become more common place. Be specific.

3. (25 points) Below is the shorthand notation for an electrochemical cell:
Pt | UO_2^{2+} (0.0150 M), U^{4+} (0.200 M), H^+ (0.0300 M) || Fe^{3+} (0.0100 M), Fe^{2+} (0.0250 M) | Pt

Here are the half reactions, written as reductions, and their standard electrode potentials.

$UO_2^{2+} + 4H^+ + 2e^- = U^{4+} + 2\ H_2O$	E° = +0.334 V vs. NHE
$Fe^{3+} + e^- = Fe^{2+}$	E° = +0.771 V vs. NHE

(a) (10 points) Draw an electrochemical cell that illustrates the cell described by the shorthand notation above. Include salt bridge, electrodes, beakers, show reaction taking place at each electrode, etc.
(b) (5 points) Which half reaction takes place at the cathode? Explain.
(c) (10 points) Is this a galvanic or electrolytic cell? Show calculations.

4. (15 points) Calculate the equilibrium constant, K_{redox}, for the following reaction that takes place in an aqueous solution. Assume that the $[H^+] = 1.50 \times 10^{-3}$ M.

$$VO^{2+} + V^{2+} + 2H^+ = 2V^{3+} + H_2O$$

Note that the half reactions are:

$V^{3+} + e- = V^{2+}$	E° = -0.255 V vs. NHE
$VO^{2+} + 2H^+ + e^- = V^{3+} + H_2O$	E° = +0.359 V vs. NHE

Equations that might be useful.

(assume that activities are equal to molar concentrations; activity coefficients are unity)

$$E = E^o + \frac{RT}{nF}\ln\frac{\text{oxidized forms}}{\text{reduced forms}} = E^o + \frac{RT}{nF}\ln\frac{[A]^a[B]^b}{[C]^c[D]^d}$$

$$E = E^o + \frac{0.0592}{n}\log\frac{[A]^a[B]^b}{[C]^c[D]^d} \text{ at } 25\,°C$$

$$-n\,F\,E_{rxn} = \Delta G$$

$$E_{rxn} = E_{cell} = E_{red} - E_{ox}$$

$$E_{cell} = E_{cathode} - E_{anode}$$

$$E_{cell} = E_{right} - E_{left}$$